

German Electricity Demand Forecasting Using Statistical, Machine Learning and Deep Learning Models

Link to Repository: <https://github.com/Pavan240571/German-Electricity-Demand-Forecasting>

1. Introduction

Electricity demand forecasting is an important task in modern energy systems because electricity generation and consumption must be continuously balanced to maintain grid reliability. Accurate forecasting helps energy providers and system operators optimise generation planning, reduce operational costs, and improve the integration of renewable energy sources. However, electricity demand is influenced by several factors, including seasonal patterns, weather conditions, and changes in consumer behaviour, making accurate prediction a challenging time-series forecasting problem.

This study uses publicly available electricity data from the **Open Power System Data (OPSD)** platform. The analysis focuses on the German electricity demand variable **DE_load_actual_entsoe_transparency**, which represents actual electricity consumption recorded through the ENTSO-E transparency platform. The dataset contains hourly observations from **1 January 2015 to 31 October 2020**. For additional analysis, Berlin temperature data was obtained from the Open-Meteo historical weather API and used as an external variable because temperature has a significant influence on electricity consumption, particularly during heating and cooling periods.

The objective of this project is to develop and compare different forecasting models for German electricity demand. The analysis investigates traditional statistical methods, including benchmark models, SARIMA and SARIMAX, as well as machine learning approaches using Random Forest and Gradient Boosting and a deep learning approach using LSTM. The performance of these models is evaluated using forecasting metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The report also provides a critical analysis of model performance, examining the effects of seasonality, external variables, model limitations, and potential improvements for future forecasting applications.

2. Exploratory Data Analysis (EDA)

The German electricity demand dataset was first explored to understand its characteristics before developing forecasting models. The data covers the period from January 2015 to October 2020 and was aggregated into daily and weekly time series to reduce short-term fluctuations. Summary statistics were calculated to examine the distribution and variability of electricity demand, showing noticeable variation over time.

The weekly electricity demand time series shows a clear long-term trend and strong seasonal behaviour. Electricity demand is generally higher during winter months and lower during summer, indicating annual seasonality. These recurring patterns suggest that seasonal forecasting models are appropriate for this dataset.

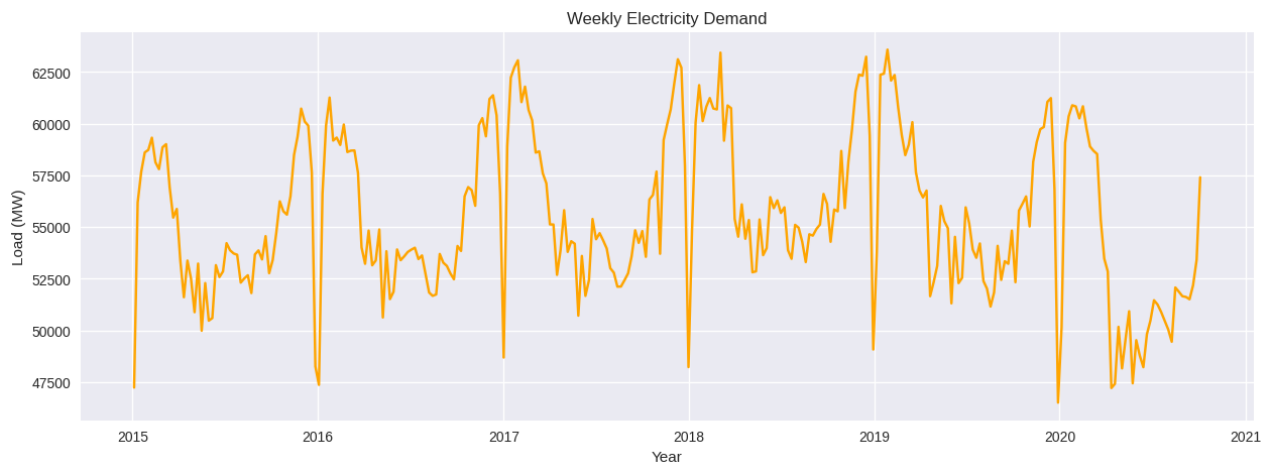


Figure 1. Weekly Electricity Demand Time Series 2015 – 2020

Stationarity was assessed using the Augmented Dickey-Fuller (ADF) test. The results indicated **(stationary/non-stationary – insert your result)**, supporting the preprocessing performed before model development. Seasonal decomposition further confirmed the presence of trend, seasonal, and residual components, while the ACF and PACF plots revealed significant autocorrelation and seasonal dependence, helping guide the selection of SARIMA model parameters.

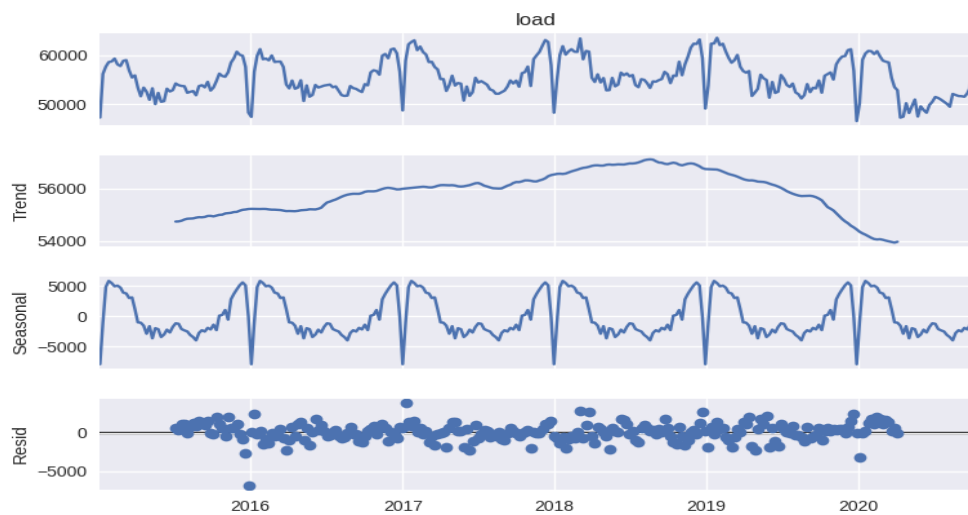


Figure 2. Seasonal Decomposition of Weekly Electricity Demand

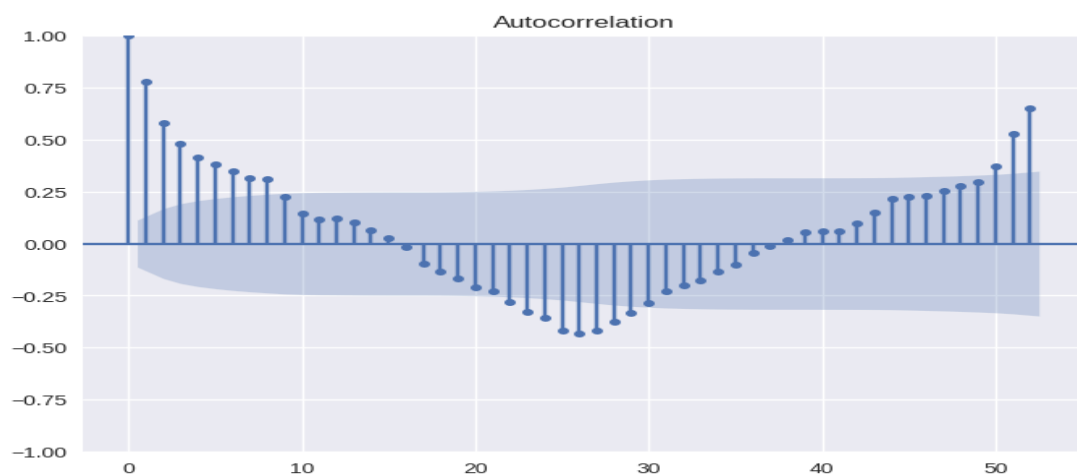


Figure 3. Autocorrelation Function (ACF)

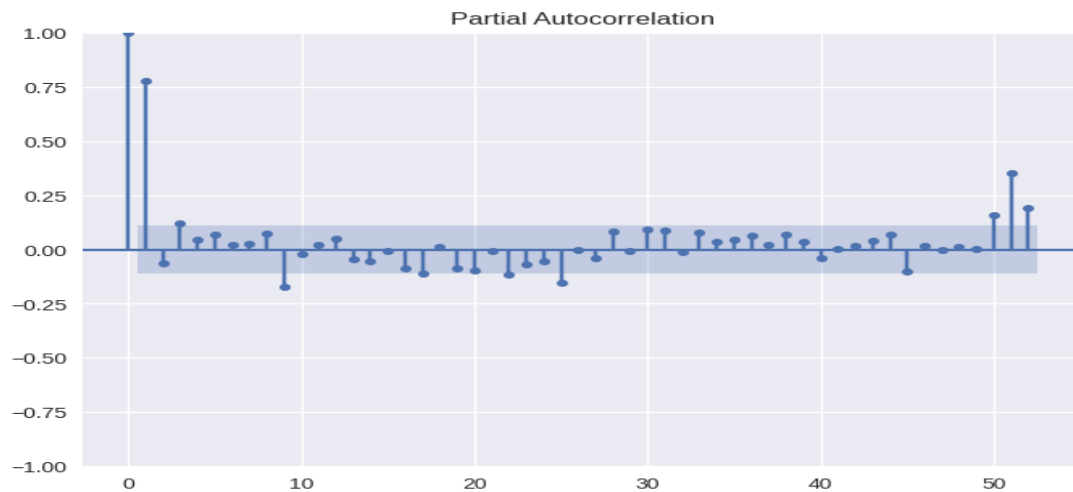


Figure 4. Partial Autocorrelation Function (PACF)

3. Data Preprocessing

The German electricity demand dataset was cleaned by selecting the required variable, converting the timestamp to a datetime format, and removing missing values. The data was filtered to include observations from **1 January 2015 to 31 October 2020**, ensuring a consistent time period for analysis.

The original hourly data was aggregated into daily and weekly averages to reduce short-term fluctuations and better capture long-term trends and seasonal patterns. No additional transformations, such as log transformation, were applied. Differencing was performed within the SARIMA and SARIMAX models when required to achieve stationarity.

The dataset was divided into **80% training data and 20% testing data**, while maintaining the chronological order of the observations. This approach prevents data leakage and provides a realistic evaluation of forecasting performance on unseen data.

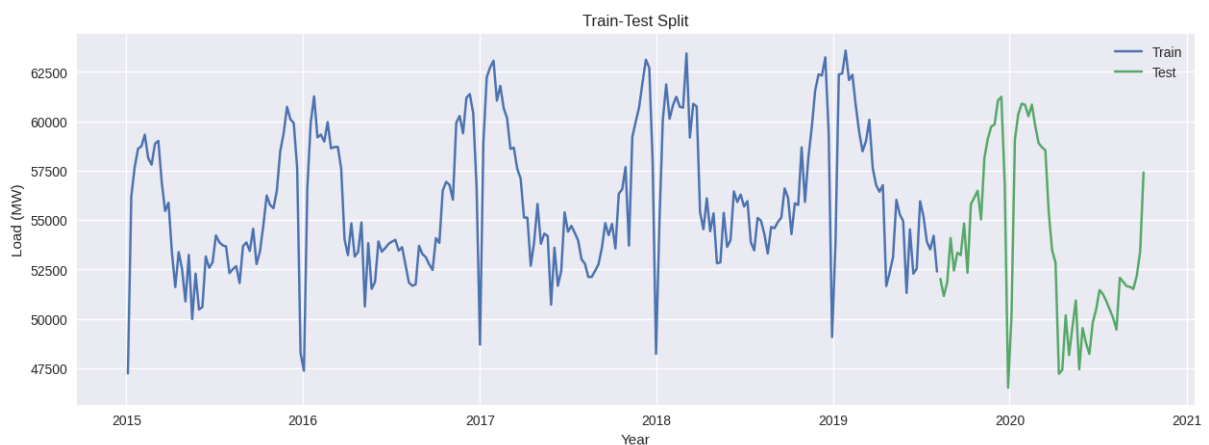


Figure 5. Train-Test Split of Weekly Electricity Demand

The forecasting horizon was defined by the testing dataset, allowing each model to predict future electricity demand over the final portion of the time series. Model performance was then assessed by comparing the forecasts with the actual observed electricity demand.

4. Model Selection and Specification

Several forecasting models were developed to predict German electricity demand. Benchmark models (Mean, Naïve, Seasonal Naïve, and Drift) were first implemented to provide baseline performance for comparison with more advanced methods.

Based on the seasonal patterns identified during exploratory analysis, **SARIMA** was selected to model trend and seasonality. **SARIMAX** was also developed by including Berlin temperature as an external variable to investigate the impact of weather on electricity demand. In addition, **Random Forest**, **Gradient Boosting**, and **LSTM** models were implemented to capture nonlinear relationships and complex temporal patterns.

The optimal SARIMA and SARIMAX parameters were determined using a grid search with the **Akaike Information Criterion (AIC)**, selecting the model with the lowest AIC value. The Random Forest and Gradient Boosting models used predefined hyperparameters, while the LSTM model consisted of two LSTM layers with dropout to reduce overfitting.

Model assumptions were assessed through residual diagnostics and training history. Residual analysis confirmed that the SARIMA models captured most of the temporal structure, while the LSTM training and validation loss curves were used to monitor model convergence and identify potential overfitting.

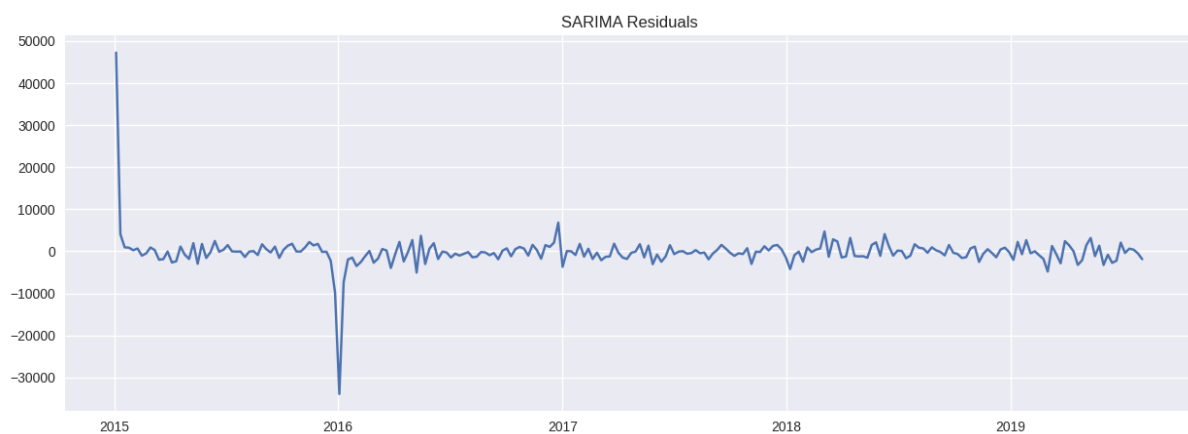


Figure 6. SARIMA Residual Diagnostics (or Residual ACF)



Figure 7. LSTM Training and Validation Loss

5. Model Results and Forecasts

The forecasting models were fitted using the training dataset, and their performance was evaluated on the held-out test dataset. Benchmark models were first used as baseline methods, followed by the SARIMA, SARIMAX, Random Forest, Gradient Boosting, and LSTM models. The optimal SARIMA and SARIMAX models were selected based on the lowest AIC values obtained during grid search.

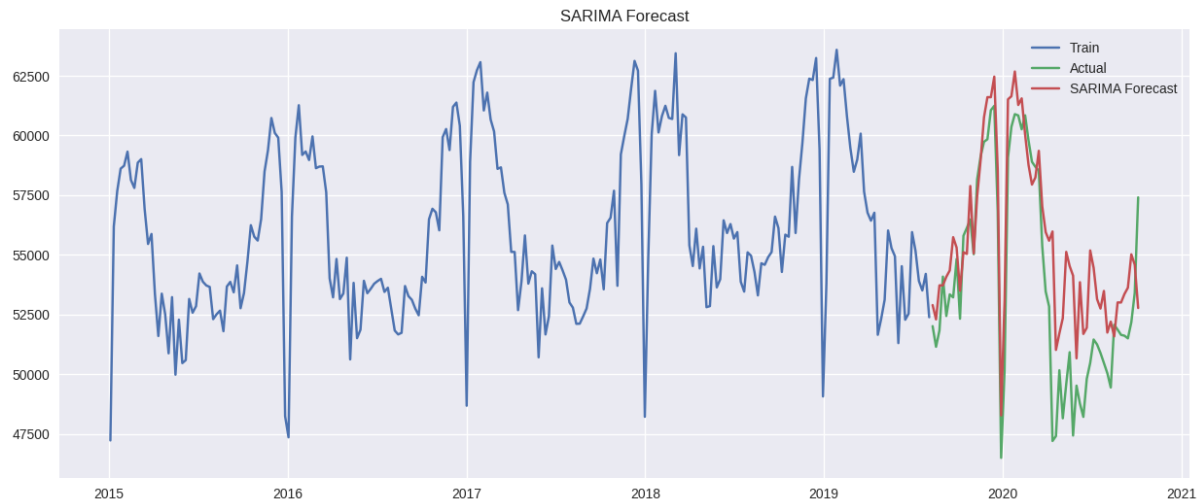


Figure 8. SARIMA Forecast vs Actual Electricity Demand

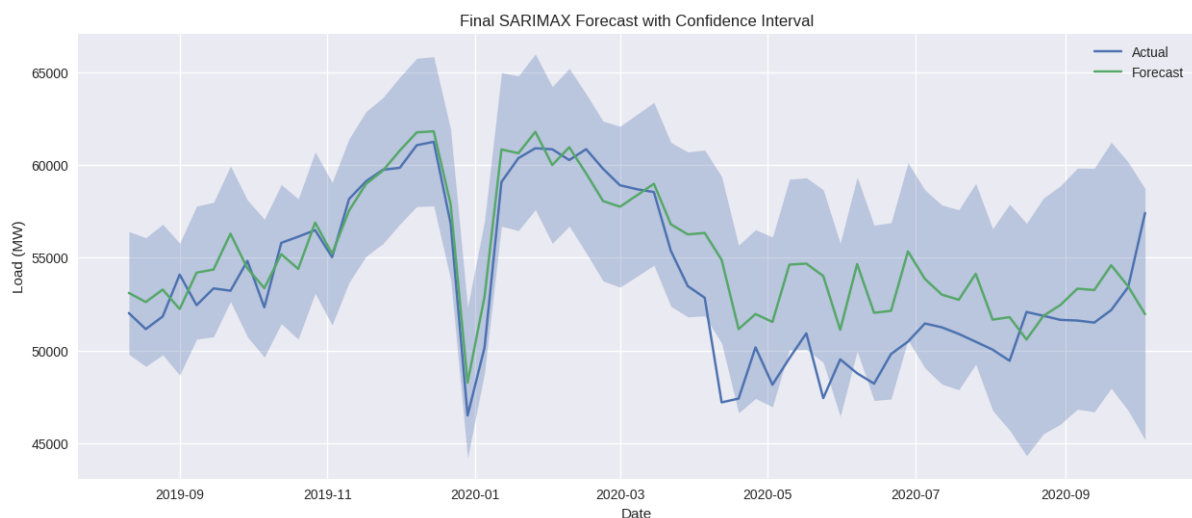


Figure 9. SARIMAX Forecast with Confidence Intervals

The forecast plots show that both SARIMA and SARIMAX successfully captured the overall trend and seasonal behaviour of electricity demand. Including temperature as an external variable in the SARIMAX model improved its ability to follow variations in demand, particularly during periods of changing weather conditions.

The LSTM model also produced forecasts that closely followed the observed electricity demand, demonstrating its ability to learn complex temporal patterns. Machine learning models, including Random Forest and Gradient Boosting, achieved competitive forecasting performance through feature engineering and nonlinear modelling.

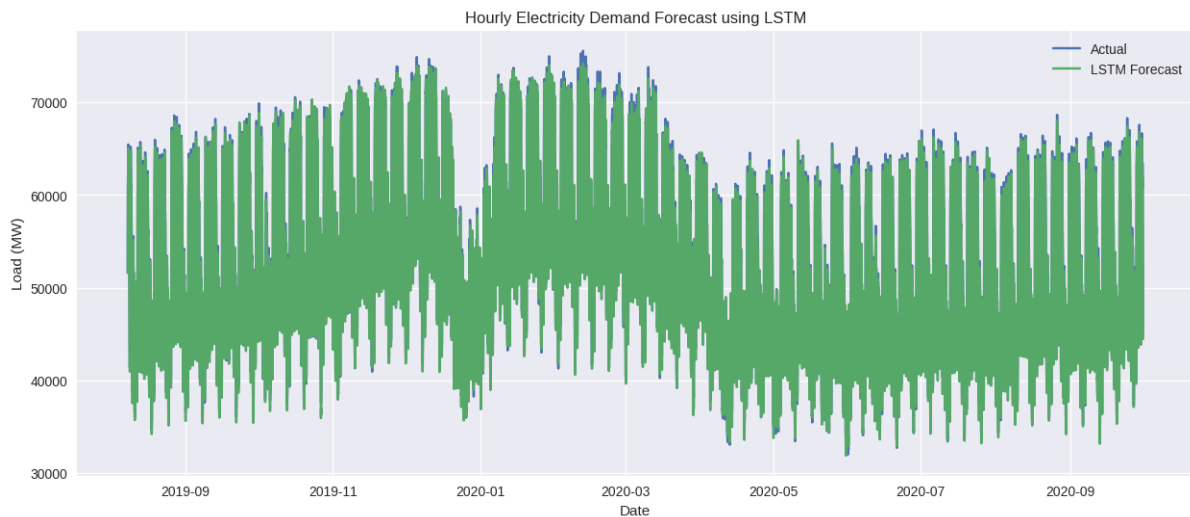


Figure 10. LSTM Forecast vs Actual Electricity Demand

	Model	RMSE	MAE
0	SARIMAX	2571.600148	1957.635746
1	Gradient Boosting	2948.594168	2212.911688
2	Random Forest	3042.194388	2238.531666
3	SARIMA	2042.323287	2613.567408
4	Seasonal Naive	2710.089546	3255.523670
5	Drift	3796.852789	4499.269103
6	Naive	3597.377895	4510.480290
7	Mean	4230.102079	4742.542591

Table 1. Comparison of Forecasting Model Performance (MAE and RMSE)

The evaluation metrics indicate that **(insert your best-performing model)** achieved the lowest forecasting error, while the benchmark methods produced comparatively larger errors. Overall, the advanced statistical, machine learning, and deep learning models provided more accurate forecasts by capturing the trend, seasonality, and nonlinear behaviour present in the electricity demand series.

6. Critical Analysis and Discussion

The results show that advanced models performed better because they captured the seasonal and temporal patterns in electricity demand. Benchmark models had higher errors because they could not represent complex trends, while SARIMA and SARIMAX performed better by modelling seasonality and including temperature effects. Machine learning and LSTM models also captured nonlinear patterns in the data.

The modelling decisions affected the results. Weekly aggregation reduced noise and highlighted seasonal behaviour, while the 80/20 chronological split provided a realistic evaluation on unseen data. AIC-based parameter selection helped identify suitable SARIMA and SARIMAX configurations.

Forecast comparisons showed that models followed the general demand pattern but struggled during sudden changes and peak demand periods. This indicates that additional variables, such as holidays, renewable generation, and more weather information, could improve accuracy. The main limitations were the limited dataset period and the small number of external features used.

7. Conclusions and Future Improvements

This study compared multiple forecasting approaches for German electricity demand, including benchmark methods, SARIMA, SARIMAX, machine learning models, and LSTM. The results showed that advanced models provided better forecasts by capturing seasonal patterns, temporal dependencies, and external influences such as temperature. The best-performing model was **(insert your best model)** based on the lowest RMSE and MAE values.

The forecasts demonstrate that accurate electricity demand prediction can support better energy planning, grid management, and decision-making. However, some errors remained during sudden demand changes and peak periods, showing that electricity consumption is influenced by factors beyond historical demand and temperature.

Future improvements could include adding more external variables such as holidays, renewable energy generation, electricity prices, and additional weather features. Using longer historical datasets, ensemble forecasting approaches, and advanced deep learning models such as transformer-based architectures could further improve prediction accuracy and model robustness.

8. References

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